Ojasva Mishra

Z Fellow | Software Engineering | AI/ML | Quantitative Research | Robotics
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Education

University of Florida Expected May 2029
B.S. Computer Engineering and Finance | GPA: 4.0/4.0 | YC Startup School 2026 | SAT 1590, National Merit *Gainesville, FL*

Experience

Drift (Antler-backed) Nov 2025 – Present
AI/Software Engineer (Founding Team) *San Francisco, CA*
- Built the **Gaussian splat pipeline** that turns photos of real spaces into walkable 3D robot simulation worlds (ROS2, Gazebo).
- Built a self-healing loop: an error-triage service clusters user-reported failures, then a repair agent patches code and reruns CI.
- Develop internal tooling and regression CI for a workspace used by **1,000+ people**, plus LLM agent evals graded by vision models.

Twinly Dec 2025 – Present
Co-Founder and CEO (Z Fellows cohort) *San Francisco, CA*
- Build AI digital clones that model a user's writing, speech, and computer behavior, then run that user's tasks in parallel.
- Engineered the Identity Runtime, a computer-use agent layer fusing accessibility APIs, browser DOM state, and multimodal vision with persistent memory; shipped on macOS, Windows, and iOS to **1,000+ active users, including paying customers**.

AlgoGators Investment Fund Sep 2026 – Present
Quantitative Research Analyst *Gainesville, FL*
- Research and backtest systematic trading signals for a ~$500K AUM student fund as 1 of 2 first-years on the quant team.
- Ran a hurricane-hedged crack-spread strategy (crude vs. refined fuels): **64% win rate over 1,200 live trades**.

Carnegie Mellon University, Xu Lab Jul 2025 – Present
Research Intern *Pittsburgh, PA*
- Design constraint-aware PID gain optimization and conformal-prediction safety filters for torque-limited robot joints.
- Validated the controllers on a Franka Emika Panda arm with Prof. Min Xu and Dr. Xiaolong Wu; **first author on 3 papers**.

EV Bots Apr 2025 – Feb 2026
AI Robotics Engineer Intern *Chester Springs, PA*
- Built the first autonomous EV-charging robot prototype and a drone-charging dock; contributed to **multiple provisional patents**.

PteroLabs Jul 2025 – Jan 2026
Software Engineering Intern *Remote*
- Engineered a UE5 UAV simulator running PX4, ArduPilot, and Betaflight autopilots in the loop (SITL), adopted at **UPenn**.

iPipeline Jun 2025 – Aug 2025
Software Development Intern *Exton, PA*
- Shipped automated testing frameworks, AI agents, and developer tooling adopted across engineering teams (Python, Docker).

Research and Publications

- Constraint-Aware Discrete-Time PID Gain Optimization for Robotic Joint Control Under Actuator Saturation. *IEEE T-RO*.
- Bounding Detector-Accepted Feedback Authority in Torque-Limited Robot Control. *IEEE ICRA 2027*.
- SCIG++: Risk-Limiting Conformal Safety Filtering for Attack-Resilient Robotic Joint Control. *IEEE/RSJ IROS 2026*.
- **UF NeuroSpine Research Team:** build nnU-Net segmentation of spine CT and MRI to automate bone-quality metrics.

Projects and Competitions

Courtside | Gator Quant Hacks 2026, Systematic Trading track | Paper | GitHub | Video Oct 2026
- Tested a pre-registered latency-arbitrage hypothesis on 13,084 Polymarket ATP/WTA tennis markets: wallets trading within 3 s of a point were **profitable after fees in all 11 out-of-sample months** (Markov fair value, deflated Sharpe, block bootstrap).
- Built a causal computer-vision ball tracker that calls points a median **162 ms before the bounce** at 119.9 fps, none wrong.

Déjà View | SwampHacks XII, 1st in AI/ML and DeepSpace tracks | Live | GitHub | Video | Devpost Oct 2026
- Trained a 1M-Gaussian splat from COLMAP poses in 11 min and exported the scene to Meta Quest, web, and Roblox.

More projects and awards
- **HIVE** (YC Transpose, 1st place): computer-vision system that maps a workspace into a task graph for its team (PyTorch).
- **VEX Robotics, 3x World Championship Competitor:** captain and lead programmer; 7 seasons of C++ autonomy and PID.

Leadership

- **FusionBots, Co-founder:** robotics nonprofit for 200+ students; wrote its 120-page open-access textbook.
- **American Rocketry Club** Founder (60+ members); **Youth Publications and Socioeconomic Forum** CTO (15+ countries).

Technical Skills

Languages: Python, C++, C#, Java, Rust, Swift, JavaScript, TypeScript, SQL, MATLAB
AI and data: PyTorch, ONNX Runtime, OpenCV, CoreML, NumPy, pandas, SciPy, LLM agents, Claude Code, Codex, Cursor
Infrastructure: Linux, Git, Docker, AWS, GCP, Slurm HPC, PostgreSQL, React, Next.js, Node.js, gRPC, CI/CD
Robotics and embedded: ROS2, Gazebo, MuJoCo, Isaac Sim, PX4, ArduPilot, Arduino, SLAM, path planning, PID control